

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.4: Linear Motion — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.4: Linear Motion
Driving Question	DRIVING QUESTION: How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line — and why does direction always matter?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Timer Race Puzzle: Launching Linear Motion	Two groups covering different path lengths arrived at the same finish point at the same time, while a third group that retraced part of the zigzag path ended up at a completely different location despite also covering significant distance.	The total length of a path and the final position of a traveller are not the same thing. The number representing distance alone does not tell us where someone ends up — direction of travel matters.	Learners attempted to explain the puzzle using their own language, identifying that moving in two opposite directions can cancel out progress toward the finish. The word displacement was introduced as the mathematical term the class will define precisely in Lesson 2.
Lesson 2 Distance, Displacement, Speed and Velocity: Giving Direction to Numbers	We watched two video clips showing that a person can travel a long path but end up close to where they started, and that the speedometer of a matatu does not tell you which direction the matatu is heading.	Distance is the total length of the path travelled (scalar). Displacement is the straight-line distance from start to finish with direction stated (vector). Speed equals distance divided by time. Velocity equals displacement divided by time and must include direction. A return journey to the starting point has zero displacement and therefore zero average velocity, even though average speed is not zero.	The timer-race puzzle is explained by the difference between displacement and distance. Group C covered a similar total distance to the other groups but a smaller displacement because they retraced steps, changing their direction and therefore their final position. The number on its own — whether distance or speed — does not tell us where someone ends up or which way they are going. Direction always matters.
Lesson 3 Velocity, Speed & Acceleration: Illustrations + Calculations	Two illustrated scenarios: a matatu accelerating from 0 to 60 km/h at a Nairobi CBD bus stop (Panel A) and a boda-boda braking from 80 km/h to 20 km/h before a speed bump on Ngong Road (Panel B). Data tables showing initial velocity (u), final velocity (v), and time (t) for each vehicle.	Velocity is the rate of change of displacement with time and is a vector quantity (has both magnitude and direction). Acceleration is the rate of change of velocity with time: $a = (v - u) / t$. Positive acceleration means the object is speeding up; negative acceleration (deceleration)	Using the matatu and boda-boda illustrated scenarios, learners computed $(v - u) / t$ from data before the teacher named it 'acceleration', allowing the formula to emerge from evidence rather than be imposed. Four graduated calculation problems (Kisumu ferry, Nairobi Express

	Computed ratios $(v - u) / t$ — one positive, one negative — extracted from real data before the formula was named.	means the object is slowing down; zero acceleration means constant velocity. Direction always matters — a boda-boda slowing down has a negative acceleration even though it is still moving forward. In the Timer Race Puzzle, the zigzag walker who retraced steps changed the direction of their velocity, which means they experienced acceleration — not just a change in position.	bus, marathon runner, unit-conversion matatu challenge) built procedural fluency and revealed that zero acceleration means constant velocity. The cumulative model diagram was updated with acceleration arrows, making the relationship between displacement, velocity, and acceleration visually explicit and anchored to the Timer Race Puzzle phenomenon.
Lesson 4 Velocity and Acceleration Problems — Practice & Sensemaking	We observed that four different Kenyan motion situations — Eliud Kipchoge's marathon run, a matatu braking on Thika Road, a mango falling near Lake Victoria, and a boda-boda overtaking on Nakuru highway — all required us to use direction (positive or negative signs) to fully describe velocity and acceleration. We noticed that the same numerical value of acceleration could mean speeding up or slowing down depending on its sign.	Velocity = displacement $\div$ time and is a vector — its sign tells us direction. Acceleration = $(v - u) \div t$ and is also a vector — positive acceleration means speeding up in the chosen direction, negative acceleration (deceleration) means slowing down, and zero acceleration means constant velocity. These two formulae are more powerful than speed and distance alone because they encode direction mathematically.	We can now explain why the Timer Race puzzle is not just a curiosity — it is a mathematical necessity. When the zigzag walker retraced steps and ended in a different place despite covering the same distance, the missing information was direction. Today's problems showed us that velocity and acceleration carry direction inside their sign. Without the sign, we cannot predict where Kipchoge ends up, whether the matatu will stop in time, or how long the mango takes to fall. Direction is not optional — it is built into the mathematics of motion.
Lesson 5 Drawing Displacement–Time Graphs: Reading Motion from a Line	Learners plotted displacement–time graphs from data tables based on the Timer Race walkers and the SGR Nairobi–Mombasa journey, identifying flat segments (stationary motion), steep segments (fast uniform motion), gentle segments (slow uniform motion), and negative-gradient segments (motion in the opposite direction, as shown by Walker C retracing her steps).	The gradient of a displacement–time graph equals the velocity of the object (gradient = $\Delta s \div \Delta t$). A positive gradient means motion away from the reference point; a zero gradient means the object is stationary; a negative gradient means motion back toward or past the reference point. This is why a displacement–time graph — unlike a speedometer or a distance value alone — can communicate both speed and direction.	Every segment of a displacement–time graph encodes information about velocity: its steepness gives the magnitude and its sign gives the direction. This is the mathematical language the Timer Race puzzle demanded — it explains why Walker C, despite covering the 'same distance,' ends up in a completely different location, and why Eliud Kipchoge's 42 km marathon displacement is far smaller than his total distance run. Direction always matters, and graphs make direction visible.
Lesson 6 Interpreting Displacement–Time Graphs	Learners read and analysed two pre-drawn displacement–time graphs: a boda-boda rider's journey in Kisumu (ferry terminal to Kondele market, with a stop and partial return) and a jogger's out-and-back run in Nairobi's Karura Forest. They observed that different segments of each graph had different shapes — some rising steeply, some flat, some sloping downward — and	Learners learned that the gradient (slope) of a displacement–time graph is equal to the velocity of the object for that segment. A steep positive gradient means high velocity in the forward direction; a shallow positive gradient means low velocity forward; a zero gradient (horizontal line) means the object is stationary; and a negative gradient means the object is moving back toward the	Learners explained that a speedometer (or a raw distance/speed number) cannot tell you the direction of travel, but the gradient of a displacement–time graph can — a negative gradient makes the return direction mathematically visible. Applied to the anchoring phenomenon, this explains why Eliud Kipchoge's marathon displacement–time graph would end near zero gradient

	recorded displacement and time values for each segment.	starting point. Because gradient encodes direction as well as speed, displacement–time graphs display velocity — a vector quantity — not merely speed.	despite 42 km of running distance, and why the retracing walker (Group C) in the Timer Race Puzzle would show a smaller final displacement than the straight-line walker on a graph — even if both graphs cover the same total time.
Lesson 7 Drawing Velocity–Time Graphs: Gradients, Acceleration, and Kipchoge's Splits	Learners plotted velocity–time graphs from tabulated data for Eliud Kipchoge's marathon training splits and a Nairobi–Thika matatu journey. They observed that different sections of the v–t graph (horizontal, rising, falling) correspond to different types of motion.	The gradient of a velocity–time graph equals the acceleration of the object ($a = \Delta v \div \Delta t$). A horizontal line means zero acceleration (uniform velocity); a positive slope means positive acceleration (speeding up); a negative slope means deceleration (slowing down). Acceleration is a vector quantity — a negative value indicates a change in direction or slowing down.	Learners used v–t graphs and gradient calculations to explain how the three Timer Race characters (straight-line walker, zigzag walker, backtracker) have different accelerations even when covering similar distances, and why knowing velocity alone is not enough — the rate of change of velocity (acceleration) and direction together determine where an object ends up.
Lesson 8 Interpreting Velocity–Time Graphs & Area Under the Graph	Learners analysed four velocity–time graph cards representing different motion scenarios (including a boda-boda journey and a matatu trip in Nairobi). They calculated gradients of each segment, identified areas of rectangles and triangles under the graphs, and multiplied axis units to determine what each area physically represents.	The gradient of a velocity–time graph equals acceleration ($\Delta v \div \Delta t$), with a negative gradient indicating deceleration. The area under a velocity–time graph equals displacement. Using these two relationships, learners calculated that the boda-boda required approximately 21 metres to stop from 30 km/h, and that a Nairobi matatu covered 96 metres total in a combined acceleration-and-braking sequence.	Learners explained why direction matters on a v–t graph by interpreting negative velocity on Graph D as the matatu reversing — connecting directly to the Timer Race Puzzle, where objects covering the same numerical distance can end up in completely different locations depending on direction. The area under the v–t graph gives the actual displacement — not just how far, but in which direction — which is exactly the richer mathematical language the driving question demands.
Lesson 9 Relative Speed — Combining Velocities When Direction Matters	Learners read an illustrated passage showing two matatus on the Nairobi–Mombasa highway — one pair travelling in the same direction and one pair travelling in opposite directions — and annotated the diagrams to compare relative speeds in each scenario.	Learners derived and applied two relative-speed rules: when two objects move in the same direction, relative speed = $v_1 - v_2$; when they move in opposite directions, relative speed = $v_1 + v_2$. They solved Kenya-contextualised problems and generalised the rule algebraically.	Direction determines whether two velocities combine by addition or subtraction. This extends the unit's central principle — that direction always matters — to situations involving two moving objects, deepening the answer to the driving question about why direction must always be specified when describing motion.
Lesson 10 Application & Cumulative Model Building: The Final Explanation	Learners returned to the Timer Race Puzzle and the full Driving Question Board (DQB), categorised accumulated evidence and questions as fully answered, partially answered, or still uncertain, and identified remaining conceptual gaps. They then received and decomposed a rich real-world problem: determining whether a matatu can	Learners integrated all unit concepts — scalar vs. vector quantities (distance/displacement, speed/velocity), the formula $v = s/t$, the acceleration formula $a = (v-u)/t$, equations of uniform motion ($s = ut + \frac{1}{2}at^2$), and the construction and interpretation of velocity-time graphs (gradient = acceleration; area under graph	The Timer Race Puzzle is now fully resolved: the same numerical value for distance, speed, or time can describe entirely different physical situations because motion in a straight line requires direction — making displacement, velocity, and acceleration vector quantities. Mathematics gives us the tools (formulae,

	safely overtake a truck on the Nakuru–Nairobi highway given specific velocity, acceleration, and distance data.	= displacement) — to solve a multi-step real-world safety problem. They verified that all three mathematical representations (algebraic, numerical, graphical) yield consistent results, experiencing mathematical coherence. They completed the DQB and delivered a final explanation tying every concept back to the anchoring phenomenon.	graphs, and models) to not only describe motion precisely but to predict and evaluate it — determining whether a highway overtake is safe, understanding why Eliud Kipchoge's displacement after a marathon is nearly zero despite running 42 km, and explaining why a matatu's speedometer alone cannot tell you where in Kenya you are headed. Direction always matters.
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